

Clones from Comonoids

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Overview

Clones in **Set**

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- 1 A (concrete) **clone** is a suboperad $\mathcal{C} \subseteq \mathcal{E}(X)$ of the endomorphism operad of a set X such that
- 1 $\mathcal{C}_n$ contains all projections $\pi_{i,n}: X^n = X \times \cdots \times X \rightarrow X$ and
 - 2 $\mathcal{C}$ is closed under the **clone product**

$$\begin{aligned} & (\varphi \bullet (\psi_1, \dots, \psi_n))(x_1, \dots, x_m) \\ &= \varphi(\psi_1(x_1, \dots, x_m), \dots, \psi_n(x_1, \dots, x_m)), \end{aligned}$$

for $\psi_1, \dots, \psi_n \in \mathcal{C}_m$, $\varphi \in \mathcal{C}_n$, and $x_1, \dots, x_m \in X$.

Uli thought he had an idea

- 1 These structures make sense as long as X is a comonoid in a braided monoidal category $(\mathbf{C}, \otimes, \mathbb{I}, \chi) \in \mathbf{BrMon}$!
- 2 The **counit** $\varepsilon_X: X \rightarrow \mathbb{I}$ defines the projections

$$\pi_{i,n} := \varepsilon_X \otimes \cdots \otimes \varepsilon_X \otimes \mathrm{id}_X \otimes \varepsilon_X \otimes \cdots \otimes \varepsilon_X.$$

- 3 Example: In **Set**, $\mathbb{I} = \{0\}$ is terminal so ε_X is unique.

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- 3 Example: In **Set**, $\mathbb{I} = \{0\}$ is terminal so ε_X is unique.
- 4 Repeated application of the **comultiplication** $\Delta_X: X \rightarrow X \otimes X$ defines the copying operation; in **Set**, this is also unique,

$$\Delta_X: X \rightarrow X \times X, \quad x \mapsto (x, x).$$

The braiding is used to bring all the comultiplied inputs into the right order to feed them into $\psi_1 \otimes \cdots \otimes \psi_n$.

Myriam steps in

- ① On closer inspection this doesn't work in general. It does if
 - ① we only consider comonoid morphisms $X^{\otimes n} \rightarrow X$,
 - ② X is cocommutative, $\Delta_X = \chi \circ \Delta_X$ ($\chi = \chi_{X \otimes X}$ braiding on $X \otimes X$), and
 - ③ χ is symmetric, $\chi \circ \chi = \text{id}_{X \otimes X}$.

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 - 3 χ is symmetric, $\chi \circ \chi = \text{id}_{X \otimes X}$.
- 2 Equivalently: The monoidal subcategory of **Com**(**C**) formed by the $X^{\otimes n}$, $n \geq 0$, is cartesian — that this is equivalent holds since a symmetric monoidal category is cartesian iff each object carries a unique cocommutative comonoid structure (Fox 1976).

A slightly different perspective on Fox's theorem

Theorem

A monoidal category $\mathbf{D}$ is cartesian iff there exists a braided monoidal category $\mathbf{C}$ such that

- 1 $\mathbf{D}$ is a monoidal subcategory of $\mathbf{Com}(\mathbf{C})$ (comonoids in $\mathbf{C}$) and*
- 2 the counit and comultiplication of all $X \in \mathbf{D}$ are morphisms in $\mathbf{D}$.*

In this case, the canonical symmetry in $\mathbf{D}$ is the restriction of the braiding in $\mathbf{C}$ and all $X \in \mathbf{D} \subseteq \mathbf{Com}(\mathbf{C})$ are cocommutative.

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Fox's version: $\mathbf{cCom}$ (cocomm. comonoids) is right adjoint to forget: $\mathbf{Cart} \rightarrow \mathbf{SymMon}$. Caveat: forget: $\mathbf{Cart} \rightarrow \mathbf{BrMon}$ doesn't have a right adjoint ($\times$ is a coproduct in $\mathbf{Cart}$ but not in $\mathbf{BrMon}$).

But sometimes it does work

- 1 Let $X \in \mathbf{C}$, $\mathbf{C} \in \mathbf{Mon}$, with $\{X^{\otimes n}\}$ not cartesian. Then $\mathcal{E}(X)$ can nevertheless be a clone; that is, it is not true that $\mathcal{E}(X)$ is cartesian as a multicategory iff $\{X^{\otimes n}\}$ is cartesian.
- 2 We will see a concrete example in $\mathbf{Com}(\mathbf{Ab}^{\mathrm{op}}) = \mathbf{Ring}^{\mathrm{op}}$.

Cartesian categories

Semicartesian categories

- 1 Let $(\mathbf{C}, \otimes, \mathbb{I})$ be a (strict) monoidal category.

Definition

$\mathbf{C}$ is **semicartesian** if $\mathbb{I}$ is terminal.

- 2 Note: For any monoidal category, the slice category over $\mathbb{I}$ is semicartesian. For example, affine spaces (vector spaces plus a linear functional on them) form a semicartesian category.

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- 2 Note: For any monoidal category, the slice category over $\mathbb{I}$ is semicartesian. For example, affine spaces (vector spaces plus a linear functional on them) form a semicartesian category.
- 3 Equivalently, there is a natural transformation $\varepsilon: \text{id}_{\mathbf{C}} \Rightarrow \mathbb{I}$ with components $\varepsilon_X: X \rightarrow \mathbb{I}$ such that $\varepsilon_{\mathbb{I}} = \text{id}_{\mathbb{I}}$ (“**uniform deletion**”).

Cartesian categories

- 1 If $(\mathbf{C}, \otimes, \mathbb{I})$ is semicartesian, we obtain two wannabe projections

$$\pi_{X,Y}^1 := \text{id}_X \otimes \varepsilon_Y : X \otimes Y \rightarrow X,$$

$$\pi_{X,Y}^2 := \varepsilon_X \otimes \text{id}_Y : X \otimes Y \rightarrow Y.$$

Definition

$\mathbf{C}$ is **cartesian** if these turn $\otimes$ into a (direct) product.

Uniform copying

- 1 In this case, we write

$$f * g: Z \rightarrow X \otimes Y$$

for the unique morphism such that

$$\pi_{X,Y}^1 \circ (f * g) = f: Z \rightarrow X, \quad \pi_{X,Y}^2 \circ (f * g) = g: Z \rightarrow Y.$$

- 2 In particular, $\Delta_X := \text{id}_X * \text{id}_X: X \rightarrow X \otimes X$ turns any object X into a cocommutative comonoid (“**uniform copying**”).

Remarks on the braiding

- 1 There is a canonical symmetry

$$\chi_{X \otimes Y} := \pi_{X,Y}^2 * \pi_{X,Y}^1: X \otimes Y \rightarrow Y \otimes X$$

and Δ_X is cocommutative (if $X, Y \in \mathbf{cCom}(\mathbf{C})$, $\mathbf{C} \in \mathbf{BrMon}$, then $X \otimes Y$ is cocommutative iff $\chi_{Y \otimes X} = \chi_{X \otimes Y}^{-1}$; Eckmann–Hilton).

- 2 Now one soon arrives at Fox's theorem: a braided monoidal category is cartesian iff

$$\text{forget}: \mathbf{cCom}(\mathbf{C}) \rightarrow \mathbf{C}$$

is an isomorphism. In particular, for every symmetric monoidal category $\mathbf{C}$, $\mathbf{cCom}(\mathbf{C})$ is cartesian.

Cartesian multicategories

Operads, multicategories, monoidal categories

- 1 In a **multicategory** (**coloured operad**) there is for any objects (colours) $X_1, \dots, X_n, Y$ a multimorphism set $\mathcal{C}(X_1, \dots, X_n; Y)$.
- 2 An operad is a multicategory with a single object (colour),

$$\mathcal{O}_n = \mathcal{O}(X, X, \dots, X; X).$$

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- 3 Example: Any monoidal category $\mathbf{C}$ defines a multicategory $\mathcal{R}(\mathbf{C})$ with the same objects via

$$\mathcal{R}(\mathbf{C})(X, \dots, Y; Z) := \mathbf{C}(X \otimes \dots \otimes Y, Z).$$

Note: All endomorphism operads $\mathcal{E}(X)$ embed into $\mathcal{R}(\mathbf{C})$.

Representable multicategories

Definition

$\mathcal{R}(\mathbf{C})$ is the multicategory **represented by $\mathbf{C}$** .

- 1 The free monoidal category functor $\mathcal{C} \mapsto \mathbf{F}(\mathcal{C})$ is right adjoint to $\mathcal{R}$. The unit of the adjunction is apparently an equivalence, $\mathbf{C} \cong \mathbf{F}(\mathcal{R}(\mathbf{C}))$ as monoidal category.

Action of finite cardinals on objects

- 1 Let $\mathbb{F}$ be the category of finite cardinals $\underline{n} = \{1, \dots, n\}$, $n \geq 0$, with all maps as morphisms.
- 2 A morphism $\underline{n} \rightarrow \underline{m}$ in $\mathbb{F}^{\text{op}}$ will be called a **selection**; we write $f[i]$ for the value of the map underlying f in $i \in \{1, \dots, m\}$.
- 3 If $\mathbf{C} \in \mathbf{Cart}$ and $\pi_{X_1, \dots, X_n}^i : X_1 \otimes \dots \otimes X_n \rightarrow X_i$ are the projections, we set

$$\pi_{X_1, \dots, X_n}^{(f)} := \pi_{X_1, \dots, X_n}^{f[1]} * \dots * \pi_{X_1, \dots, X_n}^{f[m]} : X_1 \otimes \dots \otimes X_n \rightarrow X_{f[1]} \otimes \dots \otimes X_{f[m]}.$$

Action of finite cardinals on morphisms

- 1 Given a tuple $(g_1, \dots, g_n)$ with $g_i \in \mathbf{C}(X_i, Y_i)$, we define the **substitution product**

$$f \wr (g_1, \dots, g_n) := (g_{f[1]} \otimes \dots \otimes g_{f[m]}) \circ \pi_{X_1, \dots, X_n}^{(f)},$$

which is a morphism

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which is a morphism

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- 2 Example: When $\mathbf{C} = \mathbb{F}^{\text{op}}$, this turns $\mathbf{C}$ into a cartesian **club**.

Cartesian multicategories

- 1 A **cartesian structure** on a multicategory $\mathcal{C}$ assigns to a selection $f: \underline{n} \rightarrow \underline{m}$ maps

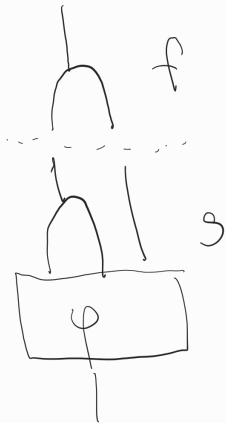
$$- \cdot f: \mathcal{C}(A_{f[1]}, \dots, A_{f[m]}; B) \rightarrow \mathcal{C}(A_1, \dots, A_n; B)$$

such that ($\circ$ in $\mathbb{F}^{\text{op}}$)

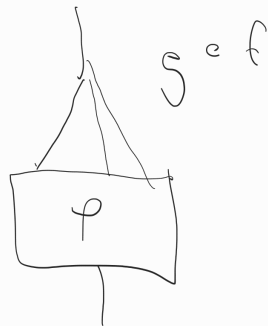
$$(\varphi \cdot g) \cdot f = \varphi \cdot (g \circ f), \quad \varphi \cdot \text{id}_{\underline{n}} = \varphi,$$

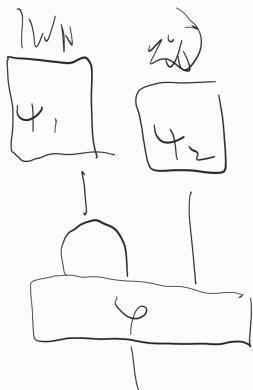
and

$$\begin{aligned} & (\varphi \cdot f) \circ (\psi_1 \cdot g_1, \dots, \psi_n \cdot g_n) \\ &= (\psi \circ (\psi_{f[1]}, \dots, \psi_{f[m]})) \cdot (f \wr (g_1, \dots, g_n)). \end{aligned}$$

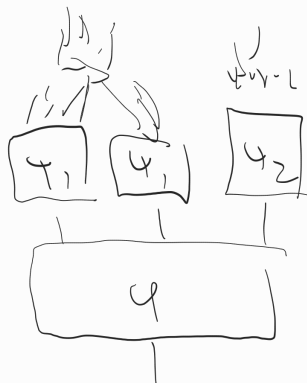


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In general, this is structure, not property

- 1 Note: A multicategory can have several cartesian structures.
- 2 However, a representable multicategory $\mathcal{R}(\mathbf{C})$ ($\mathbf{C} \in \mathbf{Mon}$) has a (unique) cartesian structure iff $\mathbf{C}$ is cartesian.

Definition

A **clone** is a cartesian operad.

As the example that comes next shows, $\mathcal{E}(X)$ can be a clone even if $\mathbf{D} = \{X^{\otimes n}\}$ and hence $\mathcal{R}(\mathbf{D})$ are not cartesian.

An example

The ambient category

- 1 Consider $\mathbf{Ab}^{\text{op}} \in \mathbf{SymMon}$ (opposite of abelian groups) and $\mathbf{C} := \mathbf{Com}(\mathbf{Ab}^{\text{op}}) = \mathbf{Ring}^{\text{op}}$ (opposite of unital associative rings).

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- 2 Aim: Find $A \in \mathbf{Ring}^{\text{op}}$ which does not generate a cartesian category $\{A^{\otimes n}\}$ yet its endomorphism operad is a clone.

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- 2 Aim: Find $A \in \mathbf{Ring}^{\text{op}}$ which does not generate a cartesian category $\{A^{\otimes n}\}$ yet its endomorphism operad is a clone.
- 3 Note: $\mathbf{cCom}(\mathbf{Ab}^{\text{op}})$ is the opposite of the category of commutative rings, that is, the category of **affine schemes** which is cartesian. The idea is to look for rings which are only mildly noncommutative.

The example

- 1 Let $q \in \mathbb{N}$ be not a square and

$$A := \mathbb{Z}\langle t, x \rangle / \langle t^2 - q, tx + xt, x^2 \rangle.$$

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- 2 If $I = Ax = xA$ is the ideal generated by x , then

$$I^2 = \emptyset, \quad A/I \cong \mathbb{Z}[\sqrt{q}],$$

so A is an **infinitesimal deformation** (or **square zero extension**) of the commutative ring $\mathbb{Z}[\sqrt{q}]$.

The clone

- 1 This ring does have quite a few automorphisms: for every $a, b \in A$ there is one with

$$t \mapsto \pm t + ax, \quad x \mapsto bx.$$

- 2 However, the commutation relations are so restrictive that there are no “extra” ring morphisms $A \rightarrow A \otimes_{\mathbb{Z}} \cdots \otimes_{\mathbb{Z}} A$, so $\mathcal{E}(A)$ is generated by automorphisms plus the projections

$$\pi_{n,i}: a \mapsto 1 \otimes_{\mathbb{Z}} \cdots \otimes_{\mathbb{Z}} 1 \otimes_{\mathbb{Z}} a \otimes_{\mathbb{Z}} 1 \otimes_{\mathbb{Z}} \cdots \otimes_{\mathbb{Z}} 1$$

and it is easily seen to be a clone.